

Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

Phu-Quy Le-Tang¹, Quang-Vu Nguyen¹, and Sunarin Chanta² *

¹ The University of Danang, Vietnam - Korea University of Information and Communication Technology

² King Mongkut's University of Technology North Bangkok
quyltp.22git@vku.udn.vn, nqvu@vku.udn.vn, sunarin.c@itm.kmutnb.ac.th

Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios. The model explicitly accounts for broken inter-hub links, multi-modal transport (land, water, and air), pre-positioned inventory, as well as adaptive lateral transshipments and reactive hub activations to recover from severe localized deficits. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a shared priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}| + |\mathcal{I}|)$. Experiments on standard benchmarks and a realistic CV-Small instance validate near-optimal accuracy against exact Pareto fronts. On the full-scale CV-Large case study under matched 20-seed run budgets, PB-NSGA uniquely achieves 100% out-of-sample feasibility on adversarial extreme scenarios and delivers a quantified cost-deprivation trade-off and a stable hub investment hierarchy for four flood-prone provinces in Central Vietnam.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1].

* Corresponding author.

As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

Humanitarian operations during this period faced significant challenges. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes “incomplete”, characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics prioritises speed and equity over cost minimisation [4]. Under uncertainty, stochastic programming and robust optimisation are the dominant paradigms [5,6], with deprivation cost—an exponential penalty on victim waiting time—capturing social welfare [7]. Hub-and-spoke structures consolidate relief flows efficiently [8]; for disaster settings, *incomplete* topologies in which infrastructure damage severs inter-hub arcs are more realistic [9]. Sangsawang and Chanta [10] established the flood-specific hub-location baseline (single-objective, deterministic, Thailand), and multi-modal extensions further address heterogeneous transport modes [11].

Two-stage stochastic programming separates pre-disaster design from post-disaster recourse [12]; robust formulations and prepositioning models outperform deterministic baselines under demand and supply uncertainty [13,14]. NSGA-II [15] is the standard multi-objective framework in humanitarian logistics, valued for parameter-free Pareto ranking and elitist selection [11,16].

Research gap and contributions. To the best of our knowledge, no existing model simultaneously addresses all four of the following: (1) incomplete hub networks with severed inter-hub arcs, (2) two-stage stochastic programming under flood uncertainty, (3) bi-objective optimisation of logistics cost *and* deprivation cost, and (4) a calibrated flood-disaster case study for Central Vietnam. Deterministic models [10] are the closest precedent: they assume full network connectivity and optimise a single cost objective, causing them to fail structurally when floods sever road links—the very scenario we target. Stochastic models [12,13] capture demand and supply uncertainty but do not model link failures in the hub sub-network, and omit the deprivation cost that captures social equity across victim groups. This paper closes all four gaps simultaneously by formulating MO-IHLNDP and solving it via PB-NSGA, a priority-based [17]

evolutionary algorithm whose encoder-decoder design efficiently handles the coupled first- and second-stage decisions of the two-stage stochastic program.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, necessitating a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo’s continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$ Set of candidate hubs (k, h), demand nodes (i), and origins (j)
 $\mathcal{S}, \mathcal{M}$ Set of disaster scenarios (s) and transportation modes (m)
 $\mathcal{V}$ Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k Fixed cost to establish hub k of the first type (before the disaster)
 c_k, κ_k Unit holding cost and capacity limit for inventory at hub k
 α, χ, M Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
 C_{uvm}, τ_{uvm} Unit transportation cost and estimated travel time from node u to v via mode m
 C_m, Q_m Unit local routing cost and transportation capacity for vehicle mode m
 γ Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F_{ks}^a	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvm}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u , and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i , and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s : $c_{jks} = \min_{m \in \mathcal{M} a_{jkm} = 1} C_{jkm}$, ($+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuitry factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s , 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s , 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s , 0 otherwise
q_k	Amount of relief items inventoried at hub k
f_{khms}	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
 \text{Min } Z_1 = & \underbrace{\sum_{k \in \mathcal{H}} F_k x_k + \sum_{k \in \mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
 & + \sum_{s \in \mathcal{S}} \pi_s \left[\underbrace{\sum_{k \in \mathcal{H}} F_{ks}^a y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \right. \\
 & \left. + \underbrace{\sum_{k \in \mathcal{H}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k \in \mathcal{H}} \sum_{i \in \mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \right] \quad (1)
 \end{aligned}$$

Objective Z_1 minimizes total expected logistics cost. Using the previously mentioned CA [18], demands are aggregated on a grid basis to pre-compute the last-mile routing cost Θ_{kis} . The first term is line-haul cost from hub k to demand i , and the second term is local detour cost inside area i :

$$\Theta_{kis} = \min_{m \in \mathcal{M} | a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} [D_{is} \cdot (e^{\lambda_{is} \cdot \Omega_{is}} - 1)] \right) \quad (2)$$

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M} | a_{ikms}=1} \tau_{kim} \right)$$

Objective Z_2 minimizes the expected value of maximum deprivation cost [7]. The exponential function of waiting time Ω_{is} heavily penalizes the model if it lets a demand node wait slightly longer than the rest, forcing the model to distribute resources evenly. The penalty function is adjusted by $\lambda_{is} = \lambda_0(1 + r_{is})$ so that highly vulnerable populations are prioritized.

To linearize Z_2 , we exploit the single-allocation property (8). Any demand i is allocated to a single hub k . Thus, the deprivation cost C_{iks}^{dep} is pre-computed:

$$C_{iks}^{\text{dep}} = D_{is} \cdot \left(e^{\lambda_{is} \cdot (\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M} | a_{ikms}=1} \tau_{kim})} - 1 \right) \quad (3)$$

By introducing a continuous auxiliary variable $W_s \geq 0$ and constraint (4) to capture the worst-case deprivation cost among all demand nodes in a given scenario s , we obtain the linearized program:

$$\begin{aligned} \text{Minimize } & Z_1, Z_2^{\text{linear}} = \sum_{s \in \mathcal{S}} \pi_s \cdot W_s \\ \text{subject to } & \text{Constraints (4) -- (18)} \\ & W_s \geq C_{iks}^{\text{dep}} \cdot z_{iks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (4)$$

2.5 Constraints

The original, complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned} \text{Minimize } & Z_1, Z_2 \\ \text{subject to } & \text{Constraints (5) -- (18)} \\ & x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (5)$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \quad (6)$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (7)$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \quad (8)$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \quad (9)$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (10)$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (11)$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (12)$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (13)$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \quad (14)$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} &\leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (15)$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (16)$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \quad (17)$$

$$q_k, f_{khms}, W_s \geq 0 \quad \forall k, h, m, s \quad (18)$$

Constraints (5)–(7) govern hub establishment and safe-zone placement; (8)–(13) enforce single-allocation via available arcs; (14)–(16) maintain flow feasibility and capacity; (17)–(18) define variable domains.

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder. The decoder is algorithm-agnostic: the same reconstruction and evaluation core is also embedded in GWO-HD and VNS-TS, so cross-algorithm differences later reflect search behavior rather than different recourse logic. We embed NSGA-II [15] rather than MOEA/D [20]: MOEA/D requires pre-specified decomposition weight vectors, presupposing the Pareto front geometry a priori; NSGA-II's crowding-distance selection preserves spread across the full Z_1 – Z_2 spectrum without weight tuning; and the encoder-decoder design is algorithm-agnostic—the *decoder* is the contribution, and NSGA-II is the dominant operator in the humanitarian logistics MOEA literature [11, 16], enabling direct comparison.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0, 1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \dots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i . The continuous vector $\mathbf{W} \in [0, 1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency ($\lambda \cdot D$) and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. Each demand i is assigned to the best reachable active hub, scored by travel time and residual capacity, starting from its anchor π_{A_i} . If all candidates are at capacity (*Pass 2*), the assignment still proceeds with a capacity violation. Only when no reachable active hub exists does the decoder open a new reactive hub as a last resort, incurring its setup cost and incrementing CV.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

With hub activations and demand assignments fixed by Steps 1–3, the supply routing in Step 4 reduces to a **minimum cost flow (MCF)** sub-problem: origins are sources, under-supplied hubs are sinks, and surplus inventory is redistributed via lateral transshipment arcs. Solving MCF exactly per scenario ensures that Z_1 is minimised to near-optimality for any decoded hub configuration, and is a key driver of PB-NSGA’s solution quality.

3.3 Genetic Operators

The initial population seeds $\mathbf{X}$ with 1 to $\lfloor 0.6|\mathcal{H}| \rfloor$ open hubs, $\mathbf{R}$ from discrete pre-positioning levels, and $\mathbf{A}, \mathbf{W}$ uniformly. Crossover (probability p_c) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX (index η_{SBX}) to $\mathbf{R}$ and $\mathbf{W}$. Mutation (base rate p_m , decaying linearly from p_m^{high} to p_m^{low} over G generations) applies bit-flip, random-reset, and polynomial perturbation to the respective segments.

Algorithm 1 Priority-Based Decoder (single scenario s)

```
1: Activate  $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1, r_{ks} \leq \chi\}$ ; set  $q_k \leftarrow R_k \kappa_k$ 
2: if  $\mathcal{H}_{\text{act}} = \emptyset$  then
3:   Force-activate  $\arg \min_k r_{ks}$  among  $\{k : X_k = 1\}$ 
4: end if
5: Score demands by urgency, isolation, and proximity; sort descending
6: for each demand  $i$  (in priority order) do
7:    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$ ; trial order  $\leftarrow \pi_{A_i}$ 
8:   Pass 1:  $k^* \leftarrow \arg \max \text{HubScore}(k)$  over first  $K$  active, reachable hubs with
       residual  $> 0$ 
9:   if Pass 1 succeeds then
10:    Assign  $i \rightarrow k^*$ ; deduct demand from residual inventory of  $k^*$ 
11:   else if any active reachable hub remains then
12:    Pass 2: assign  $i$  to first such hub (capacity violation allowed)
13:   else
14:    Reactive fallback: open nearest safe inactive hub; add  $F_{ks}^a$ ;  $\text{CV}^s += 1$ 
15:   end if
16: end for
17: Route supply from origins to hubs and redistribute surplus via MCF on the hub
    sub-network.
18: Accumulate  $Z_1^s, Z_2^s$ 
19: return  $(Z_1^s, Z_2^s, \text{CV}^s)$ 
```

Survival follows NSGA-II elitist constrained dominance [15]: feasibility first, then non-dominated rank, then crowding distance with Hamming tiebreaking in $\mathbf{X}$ -space. Rank-1 stagnation triggers increased tournament size and $\mathbf{W}$ -mutation, following the adaptive elitist strategy of [21].

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an 11th Gen Intel Core i5-1135G7 @ 2.40 GHz and 16 GB RAM. Algorithm parameters are fixed for all runs: $N = 200$, $G = 300$ (CV-Small) / 500 (CV-Large), $p_c = 0.98$, $p_m^{\text{high}} = 0.40$, $p_m^{\text{low}} = 0.10$ (linear decay), $\eta_{\text{SBX}} = 1.5$, $\eta_{\text{poly}} = 8$. The parameters are chosen via a grid search process. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean $\pm$ std. We use **Hypervolume (HV)** [22] and **IGD⁺** [23] as performance metrics, computed against the combined non-dominated reference front pooled across all algorithms.

4.2 Dataset Description

Central Vietnam Dataset. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang

Ngai. **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates map to real administrative centroids, confirmed logistics facilities, and vital supply origins. Scenario risks and road disruptions are calibrated using a multi-criteria flood vulnerability score [24]. Three transport modes – trucks, motorboats, and helicopters – are considered, with helicopter links capped at 15% of active routing connections per scenario.

Scenario generation via SAA [25]. To improve coverage of plausible flood conditions, scenarios are drawn from a profile bank spanning mild, severe, and extreme flood regimes with varying epicenter count, intensity, road disruption rate, and circuitry coefficient. The CV-Large case study further validates the first-stage decisions $(\hat{x}, \hat{q})$ on an independent out-of-sample (OOS) set of $N' = 10$ adversarial scenarios, disjoint from training, to assess generalization under catastrophic conditions. Because this OOS holdout is deliberately extreme-only, we treat feasibility as the primary robustness metric and interpret large OOS increases in Z_2 as stress amplification under adversarial conditions rather than ordinary prediction error.

Scenario-count sensitivity. Figure 1 reports a scenario-count sensitivity study on CV-Small ($K = 10$ replications, exact enumeration). The standard deviation of Z_2 contracts from $\pm 110k$ ($N = 3$) to $\pm 30k$ ($N = 30$), confirming SAA variance reduction with scenario coverage; HV std decreases from 0.25 to 0.07. The transient peak at $N = 5$ reflects the high variance of small uncalibrated draws from an extreme-heavy pool. Our historically calibrated three-profile design eliminates this sampling uncertainty by construction, achieving stability comparable to $N \approx 10$ unstratified scenarios while halving training overhead.

4.3 Experiment 1: Baseline Comparison

We evaluate PB-NSGA against four baselines on the **CV-Small** instance. The **Greedy Heuristic** simulates ad-hoc decision-making: it constructs a council of managers with different cost-safety weight profiles, each greedily selecting hubs and assigning demands by nearest-neighbour logic, then returns their combined non-dominated solutions. **VNS-TS** [10] is a Variable Neighbourhood Search with Tabu Search, adapted from the flood relief literature as a strong meta-heuristic baseline. **GWO-HD** [11] is a Grey Wolf Optimizer with Hamming Distance moves, representing a recent meta-heuristic designed for discrete hub-location problems. The **MILP** solver [26] applies the Adaptive Weighted Sum method [27] to systematically sweep the objective space and produces the reference Pareto front for CV-Small.

Table 1 reveals a clear performance hierarchy. The MILP solver defines the reference Pareto front for CV-Small, establishing the quality ceiling at $HV = 0.997$. The Greedy heuristic runs near-instantly but remains far from the front. VNS-TS and GWO-HD both achieve competitive HV (≈ 0.78) with VNS-TS requiring 60 s and GWO-HD only 0.6 s. PB-NSGA closes the gap to MILP substantially ($HV = 0.925$, $IGD^+ = 0.039$) in 1.1 s, demonstrating that the priority-based decoder effectively navigates the combinatorial structure of MO-IHLNDP.

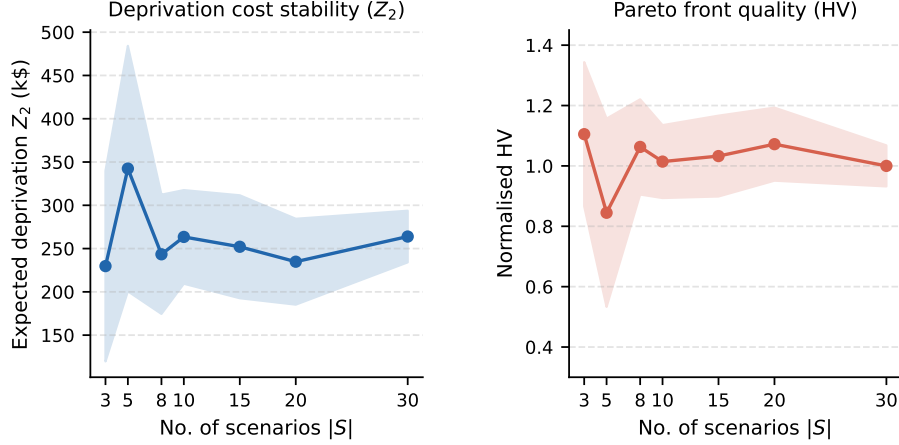


Fig. 1. Scenario-count sensitivity on CV-Small ($K = 10$ replications per N , exact bb-solver enumeration). Shaded bands show ± 1 std across independent random sub-samples from a 50-scenario balanced pool. The std of Z_2 (left) and normalised HV (right) both follow an overall decreasing trend with N , confirming that scenario-space variance contracts as coverage increases. Our fixed three-profile design achieves the stability of $N \approx 10$ random draws through deliberate stratification.

4.4 Experiment 2: Case Study on Central Vietnam

We apply PB-NSGA, GWO-HD, and VNS-TS to the full-scale **CV-Large** instance ($|I| = 100$, $|\mathcal{H}| = 20$), where exact solvers become intractable. All other parameters follow Sec. 4.1.

Figure 2 shows the Pareto fronts for both instances. On CV-Small (left panel), PB-NSGA closely tracks the MILP true front across the full cost-deprivation range; GWO-HD eventually converges to a similar region but with more scattered solutions; VNS-TS only finds the cost-efficient tail, missing the deprivation-minimizing end entirely. On CV-Large (right panel, power scale), the trade-off space spans nearly an order of magnitude in Z_1 . PB-NSGA concentrates its solutions in the low-cost, low-deprivation region (Z_1 : 4.3M–5M, Z_2 : 41K–45K), defining the efficient low-cost frontier. GWO-HD extends the front rightward, maintaining competitive $Z_2 \approx 44$ K at the cost of substantially higher Z_1 (up to 26M). VNS-TS is largely confined to the high-cost tail, with solutions reaching $Z_1 = 81$ M and $Z_2 = 150$ K.

On CV-Large across 20 matched runs per algorithm, PB-NSGA achieves $HV = 1.000 \pm 0.006$ and $IGD^+ = 0.000 \pm 0.001$ in 8.7s on average. GWO-HD reaches $HV = 0.982 \pm 0.007$ in 4.8s, while VNS-TS improves to $HV = 0.823 \pm 0.038$ but still requires 180.6s per run – a $20.7\times$ runtime disadvantage relative to PB-NSGA. PB-NSGA therefore combines the best frontier quality with sub-10-second runtime on the full-scale case study.

Table 1. Baseline comparison on CV-Small. HV and IGD^+ are normalized w.r.t. the combined non-dominated reference front across all algorithms. Best values in **bold**.

Algorithm	HV (norm.) $\uparrow$	IGD^+ $\downarrow$	CPU Time (s) $\downarrow$
Greedy Heuristic	0.292	0.333	< 0.1
VNS-TS	0.777	0.117	60.0
GWO-HD	0.780	0.152	0.6
MILP (Adaptive WS)	0.997	0.005	4.9
PB-NSGA (Ours)	0.925	0.039	1.1

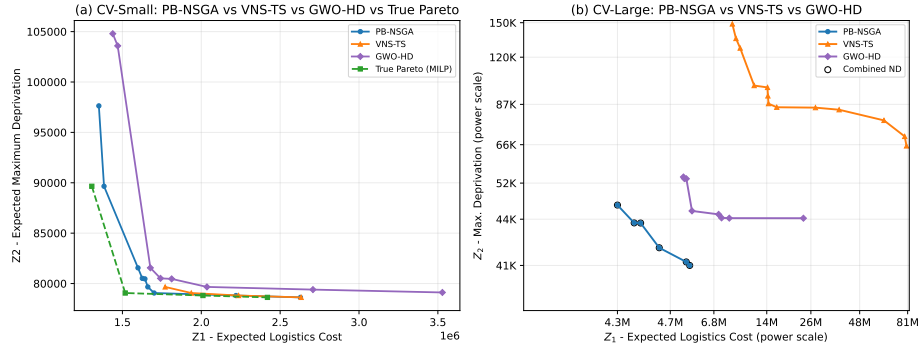


Fig. 2. Combined Pareto fronts for PB-NSGA, GWO-HD, and VNS-TS. **Left (CV-Small):** all three algorithms compared against the MILP true Pareto front. **Right (CV-Large, power scale):** combined non-dominated front across 20 matched runs for each algorithm. Z_1 : expected logistics cost; Z_2 : expected maximum deprivation cost.

These robustness results are especially informative because PB-NSGA, GWO-HD, and VNS-TS all use the same decoder and evaluation core. Across 20 matched seeds, all three algorithms achieve 100% feasibility on the SAA-100 validation set, confirming that the decoder itself is robust. On the adversarial OOS set of 10 extreme scenarios, PB-NSGA is the only method maintaining perfect feasibility across all 20 seeds, whereas GWO-HD and VNS-TS remain highly feasible at 96.4% (95% CI [93.4%, 98.8%]) and 97.8% (95% CI [96.2%, 99.2%]). The performance gap therefore comes from search quality and search efficiency, not from privileged access to a different recourse engine.

Figure 3 visualizes the knee-point solution ($Z_1 = \$4.6\text{M}$, $Z_2 = 43,453$) across three scenarios. The eight planned hubs (H0, H1, H3, H11, H14, H15, H16, H19) remain active with identical fill levels across all scenarios; no reactive hub is activated and no lateral transshipment occurs. The absence of reactive activations is not a model artifact but an optimal structural property of dense networks: with 20 candidate hubs at commune resolution, extensive pre-positioning at eight locations mathematically dominates the marginal cost of incurring reactive setup fees (\$51K–\$109K per scenario). The reactive recourse mechanism (Algorithm 1, Pass 3) correctly remains dormant here; it actively triggers in sparser topologies

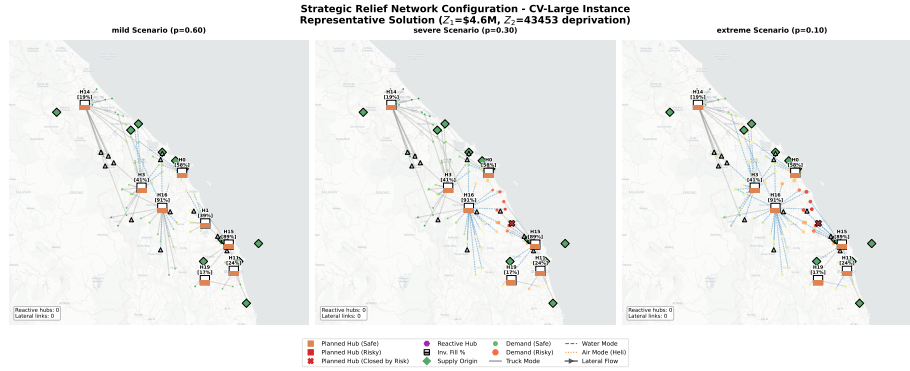


Fig. 3. Strategic relief network for a representative solution on CV-Large ($Z_1 = \$4.6M$, $Z_2 = 43,453$) across three probabilistic scenarios: mild ($p = 0.60$), severe ($p = 0.30$), and extreme ($p = 0.10$). Hub inventory fill levels are annotated in brackets. No reactive hub is activated in any scenario. Line styles indicate transport mode: solid gray (truck), blue dashed (water), orange dashed (helicopter).

such as CV-Small under severe and extreme scenarios, confirming the mechanism’s correctness. The principal adaptation across scenarios is modal rather than structural: helicopter routes displace truck links in the extreme scenario to reach nodes severed by landslide.

H16 (91%) and H15 (89%) carry the heaviest pre-positioned loads, anchoring coverage for the central and southern coastal corridor. The primary adaptation across scenarios is modal: mild and severe conditions are served by a mix of truck and watercraft links, while the extreme scenario activates helicopter routes (orange dashed) to reach high-risk demand nodes that become inaccessible by road and water.

As a sensitivity check on training-set size, we compare the calibrated three-scenario design against a tier-balanced $N = 10$ training set using 20 PB-NSGA runs each. The calibrated $N = 3$ design achieves significantly better OOS feasibility (1.000 vs. 0.931; $\Delta = -0.069$, 95% CI $[-0.128, -0.029]$) at one-third the training burden. By contrast, the mean OOS ΔZ_2 difference is not significant ($\Delta = -437\%$, 95% CI $[-1865\%, +1969\%]$) because the $N = 10$ design is highly right-skewed: even after excluding five BIG_M sentinel evaluations, one seed still exhibits a genuine catastrophic OOS failure (22,037% mean OOS ΔZ_2). We therefore treat feasibility as the primary robustness conclusion and interpret the $N = 10$ result as evidence that uniform-probability random training carries substantially higher robustness variance under adversarial extreme-only stress.

Decision support. Hub selection frequency across all 20 CV-Large runs (128 Pareto solutions) reveals a clear investment hierarchy: Quang Ngai Port (H15) and Phuoc Son Helipad (H16) appear in every solution (100%), followed by Huong Viet Depot (H14, 82.8%), Da Nang Airport (H0, 77.3%), Tam Ky Logistics Hub (H1, 67.2%), Binh Son Warehouse (H9, 59.4%), and Dong Giang

Station (H3, 57.8%). H1’s risk surges from 0.46 (mild) to 0.94 (severe), exceeding $\chi = 0.7$ and forcing deactivation under high-severity scenarios; authorities should treat it as a conditional asset and pre-stage contingency inventory at H14 (risk ≤ 0.13 across all scenarios). Under the extreme scenario, road disruption enforces a modal shift to watercraft and helicopter routes; pre-equipping H15 and H16 with marine terminals and maintaining rotary-wing access at H0 delivers asymmetric resilience at lower cost than redundant infrastructure.

5 Conclusion and Future Work

This paper makes three contributions to humanitarian logistics under uncertainty. First, we formulate MO-IHLNDP, a two-stage stochastic bi-objective model that jointly captures incomplete hub networks, multi-modal transport, deprivation cost, and pre-positioned inventory for flood disaster relief. Second, we propose PB-NSGA, a priority-based NSGA-II with an MCF-backed decoder that efficiently navigates the coupled first- and second-stage decisions. Because the same decoder and evaluation core is also embedded in GWO-HD and VNS-TS, the matched 20-seed CV-Large study isolates search from reconstruction: all three methods remain highly feasible under adversarial OOS stress, but PB-NSGA alone maintains 100% OOS feasibility while delivering the best Pareto quality in under 10s. Third, applied to four flood-prone provinces in Central Vietnam, the joint framework delivers a practical decision-support tool: a quantified cost-deprivation trade-off, a stable hub investment hierarchy anchored by the Quang Ngai Port Hub and Phuoc Son Helipad, explicit identification of resilience gaps for contingency planning, and evidence that the calibrated three-scenario design is more reliable than a tier-balanced $N = 10$ alternative at one-third the training burden.

Limitations and future work. The model is validated on a synthetic instance calibrated from administrative and geographic data; field validation against actual disaster records is necessary before operational deployment. Three directions are prioritised: integrating real-time sensor and satellite feeds into a live decision-support ecosystem; extending the model to explicitly penalize hub processing degradation under high-risk conditions, bridging scenario-based planning and tail-risk management; and scaling PB-NSGA to hour-long runtime budgets with continuous convergence guarantees for large-scale regional deployments.

References

1. Center for Disaster Philanthropy, “2025 southeast asia severe storms,” December 2025.
2. Voice of Vietnam, “(translated) natural disasters in 2025: Fierce, abnormal, causing heavy losses in humans and property,” December 2025. VOV.VN.
3. N. Altay and W. G. Green III, “Or/ms research in disaster operations management,” *European journal of operational research*, vol. 175, no. 1, pp. 475–493, 2006.

4. L. N. Van Wassenhove, "Humanitarian aid logistics: supply chain management in high gear," *Journal of the Operational research Society*, vol. 57, no. 5, pp. 475–489, 2006.
5. Z. Dönmez, B. Y. Kara, Ö. Karsu, and F. Saldanha-da Gama, "Humanitarian facility location under uncertainty: Critical review and future prospects," *Omega*, vol. 102, p. 102393, 2021.
6. C. Boonmee, M. Arimura, and T. Asada, "Facility location optimization model for emergency humanitarian logistics," *International journal of disaster risk reduction*, vol. 24, pp. 485–498, 2017.
7. J. Holguín-Veras, N. Pérez, M. Jaller, L. N. Van Wassenhove, and F. Aros-Vera, "On the appropriate objective function for post-disaster humanitarian logistics models," *Journal of Operations Management*, vol. 31, no. 5, pp. 262–280, 2013.
8. J. F. Campbell, "Integer programming formulations of discrete hub location problems," *European Journal of Operational Research*, vol. 72, no. 2, pp. 387–405, 1994.
9. S. A. Alumur, B. Y. Kara, and O. E. Karasan, "The design of single allocation incomplete hub networks," *Transportation Research Part B: Methodological*, vol. 43, no. 10, pp. 936–951, 2009.
10. O. Sangsawang and S. Chanta, "Capacitated single-allocation hub location model for a flood relief distribution network," *Computational Intelligence*, vol. 37, no. 1, pp. 345–374, 2020.
11. C. Li, P. Han, M. Zhou, and M. Gu, "Design of multimodal hub-and-spoke transportation network for emergency relief under covid-19 pandemic: A meta-heuristic approach," *Applied Soft Computing*, vol. 133, p. 109925, 2023.
12. S. Tofghi, S. A. Torabi, and S. A. Mansouri, "Humanitarian logistics network design under mixed uncertainty," *European journal of operational research*, vol. 250, no. 1, pp. 239–250, 2016.
13. M. Yahyaei and A. Bozorgi-Amiri, "Robust reliable humanitarian relief network design: an integration of shelter and supply facility location," *Annals of Operations Research*, vol. 283, pp. 897–916, 2019.
14. F. S. Salman and E. Yücel, "Emergency facility location under random network damage: Insights from the istanbul case," *Computers & Operations Research*, vol. 62, pp. 266–281, 2015.
15. K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: Nsga-ii," *IEEE transactions on evolutionary computation*, vol. 6, no. 2, pp. 182–197, 2002.
16. Y. Zhou, J. Liu, Y. Zhang, and X. Gan, "A multi-objective evolutionary algorithm for multi-period dynamic emergency resource scheduling problems," *Transportation Research Part E: Logistics and Transportation Review*, vol. 99, pp. 77–95, 2017.
17. M. Gen, F. Altiparmak, and L. Lin, "A genetic algorithm for two-stage transportation problem using priority-based encoding," *OR Spectrum*, vol. 28, no. 3, pp. 337–354, 2006.
18. C. F. Daganzo, *Logistics systems analysis*. Springer, 2005.
19. P. B. Mirchandani and R. L. Francis, *Discrete Location Theory*. New York: Wiley-Interscience, 1990.
20. Q. Zhang and H. Li, "MOEA/D: A multiobjective evolutionary algorithm based on decomposition," *IEEE Transactions on Evolutionary Computation*, vol. 11, no. 6, pp. 712–731, 2007.
21. Y. Liang and K.-S. Leung, "Genetic algorithm with adaptive elitist-population strategies for multimodal function optimization," *Applied Soft Computing*, vol. 11, no. 2, pp. 2017–2034, 2011.

22. E. Zitzler and L. Thiele, "Multiobjective evolutionary algorithms: a comparative case study and the strength Pareto approach," *IEEE Transactions on Evolutionary Computation*, vol. 3, no. 4, pp. 257–271, 1999.
23. H. Ishibuchi, H. Masuda, Y. Tanigaki, and Y. Nojima, "Modified distance calculation in generational distance and inverted generational distance," in *Evolutionary Multi-Criterion Optimization*, vol. 9019 of *Lecture Notes in Computer Science*, pp. 110–125, Springer, 2015.
24. M. T. Nguyen, Z. Sebesvari, M. Souvignet, F. Bachofer, A. Braun, M. Garschagen, U. Schinkel, L. E. Yang, L. H. Nguyen, V. Hochschild, A. Assmann, and M. Hagenlocher, "Understanding and assessing flood risk in Vietnam: Current status, persisting gaps, and future directions," *Journal of Flood Risk Management*, vol. 14, no. 2, p. e12689, 2021.
25. A. Shapiro, "Monte carlo sampling methods," *Handbooks in operations research and management science*, vol. 10, pp. 353–425, 2003.
26. L. Perron and V. Furnon, "Or-tools," 2025.
27. I. Y. Kim and O. L. De Weck, "Adaptive weighted-sum method for bi-objective optimization: Pareto front generation," *Structural and multidisciplinary optimization*, vol. 29, no. 2, pp. 149–158, 2005.